

数值分析第十七次作业

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1. 定义离散正弦变换 (DST) 为

$$F_k = \sum_{j=1}^{N-1} f_j \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leq k \leq N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \sum_{k=1}^{N-1} F_k \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leq j \leq N-1.$$

解答.

□

2. 定义离散余弦变换 (DCT) 为

$$F_k = \frac{1}{2}(f_0 + (-1)^k f_N) + \sum_{j=1}^{N-1} f_j \cos\left(\frac{jk\pi}{N}\right), \quad \forall 0 \leq k \leq N,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \left(\frac{1}{2}(F_0 + (-1)^j F_N) + \sum_{k=1}^{N-1} F_k \cos\left(\frac{jk\pi}{N}\right) \right), \quad \forall 0 \leq j \leq N.$$

解答.

□

3. 定义离散半波余弦变换 (Discrete Quarter-Wave Cosine Transform) 为

$$F_k = \sum_{j=0}^{N-1} f_j \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right), \quad \forall 0 \leq k \leq N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N} \left(\frac{1}{2}F_0 + \sum_{k=1}^{N-1} F_k \cos\left(\frac{(j + \frac{1}{2})k\pi}{N}\right) \right), \quad \forall 0 \leq j \leq N-1.$$

解答.

□

4. 通过离散余弦变换 (DCT) 推导矩阵

$$A = \begin{pmatrix} -1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -1 \end{pmatrix}_{(N+1) \times (N+1)}$$

的特征分解形式.

解答.

□